Flexibility from buildings

RES4BUILD project - Heat pumps & hot water buffers

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ATHENS Flexibility Course – 17/03/2026

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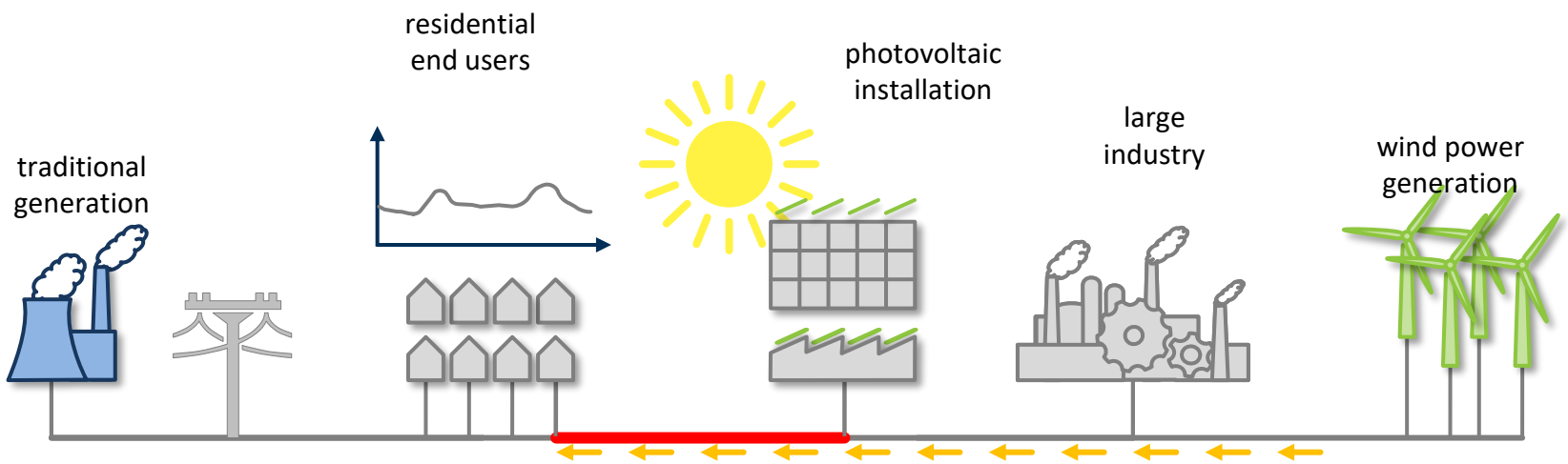
Flexibility from buildings, why?

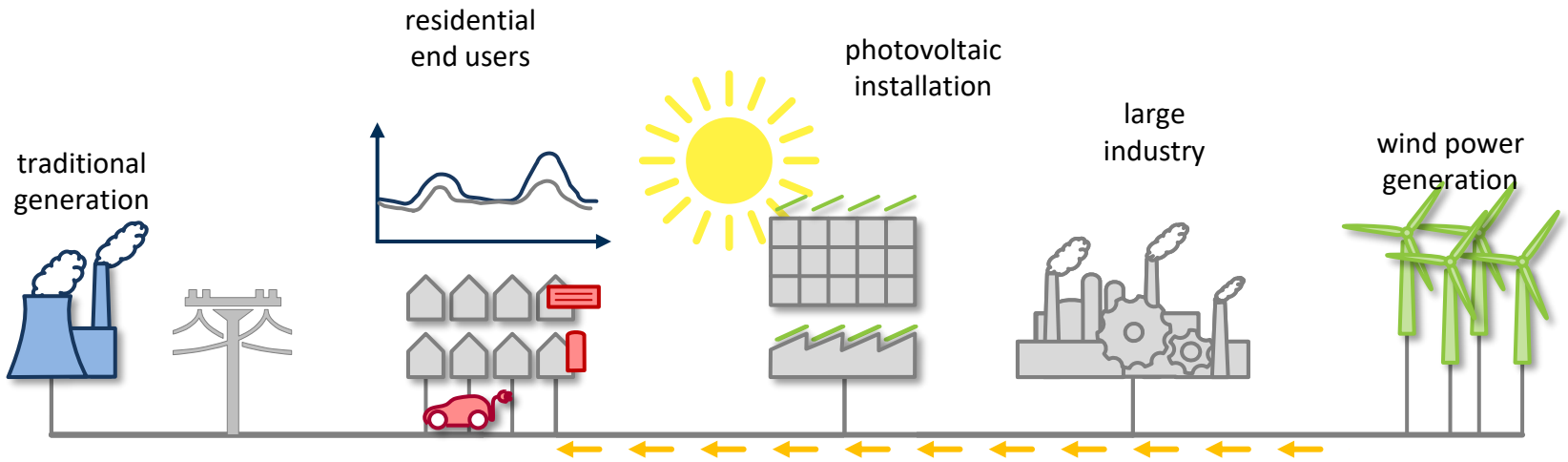


High integration of intermittent renewable energy sources in to the (distribution) grid significantly affects grid operation

Mismatch of energy demand & supply

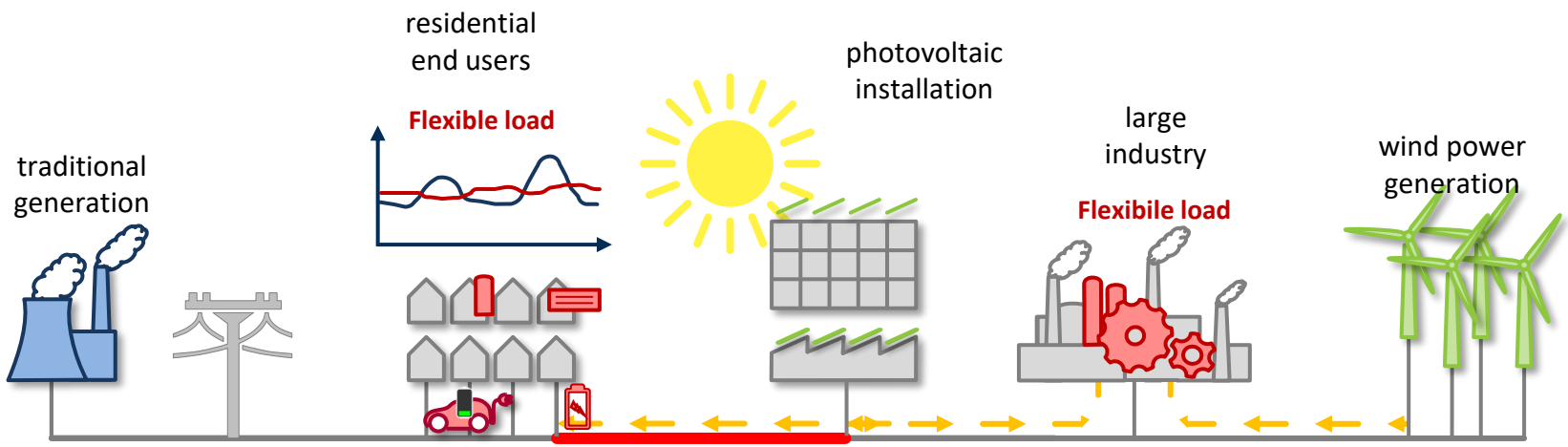
- Gas-fired power plants for balancing
- Voltage fluctuations & congestion issues

Flexibility from buildings, why?



- Electrification of heating & transportation
  - Alteration of energy demand curve → increase in peak consumption

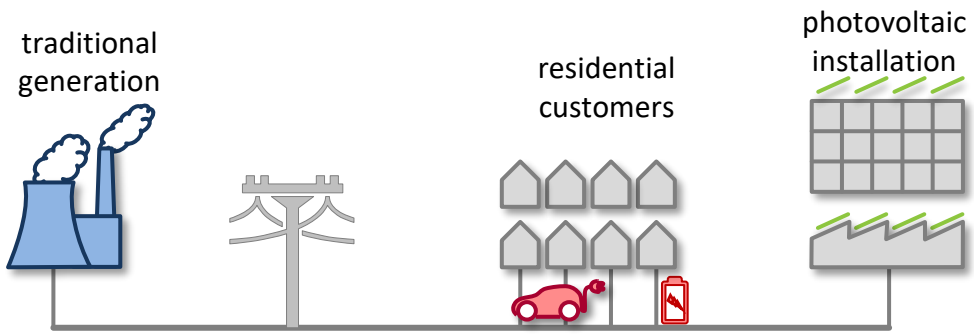
Flexibility from buildings, why?



- Exploit **flexibility** potential of **controllable devices**

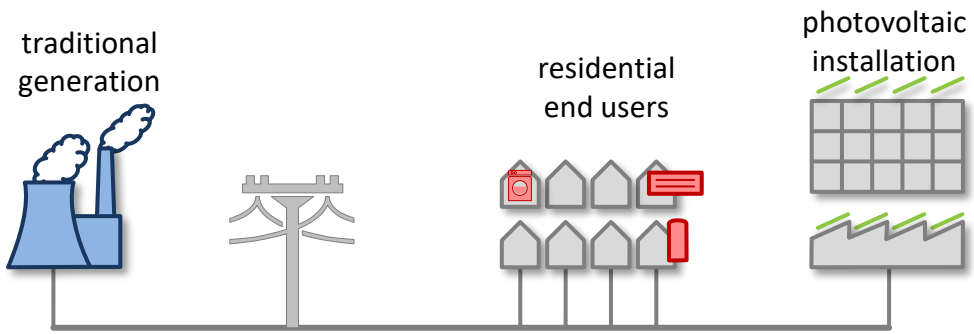
"**Flexibility** refers to the **power increase or decrease** possible within the **functional limits** of **controllable devices** & **end user comfort limits**, combined with **how long** these changes can be sustained"

Flexibility from buildings, how much?



- Battery for storage of local energy generation
  - Battery capacity & charging/discharging limits
- Battery pack in electric vehicles
  - Number of hours by which EV charging can be delayed while meeting user deadline
  - Capacity & max charging rate of the battery

Flexibility from buildings, how much?



- White appliances: dish washers, washing machines, …
  - Number of hours end user can delay appliance operation
- Space & water heating: heat pumps, hot water tanks, …
  - Flexibility attributed to thermal storage capacity of the building & the water boiler
  - Amount of energy & duration by which heat pump's consumption can be altered

Flexibility from buildings - RES4BUILD

Renewables for clean energy in buildings in a future power system

- Integrated **renewable energy-based** solutions for decarbonising **energy use** in **buildings**
- Project duration – 4 years
  - 15 partners

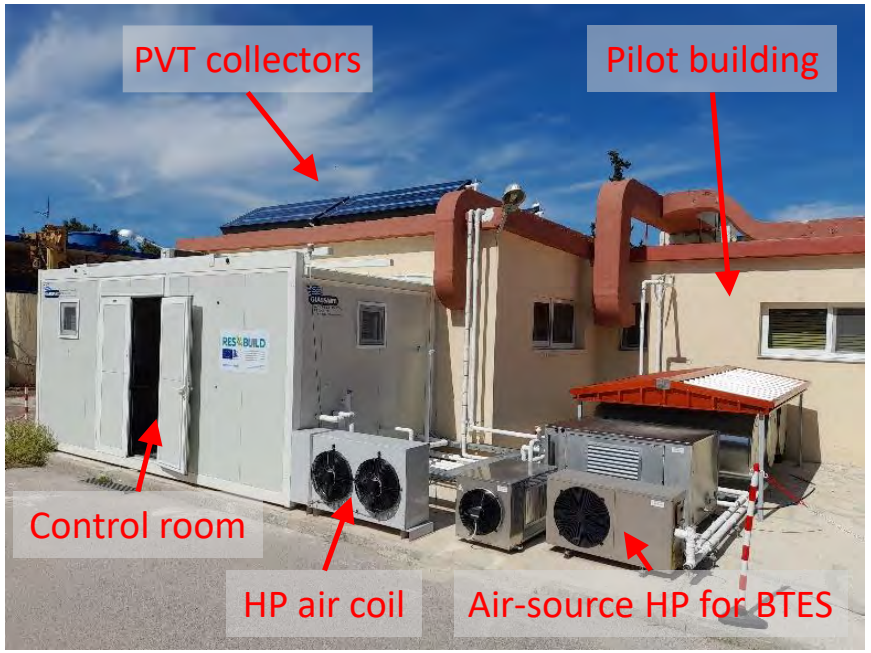
Flexibility from buildings - RES4BUILD

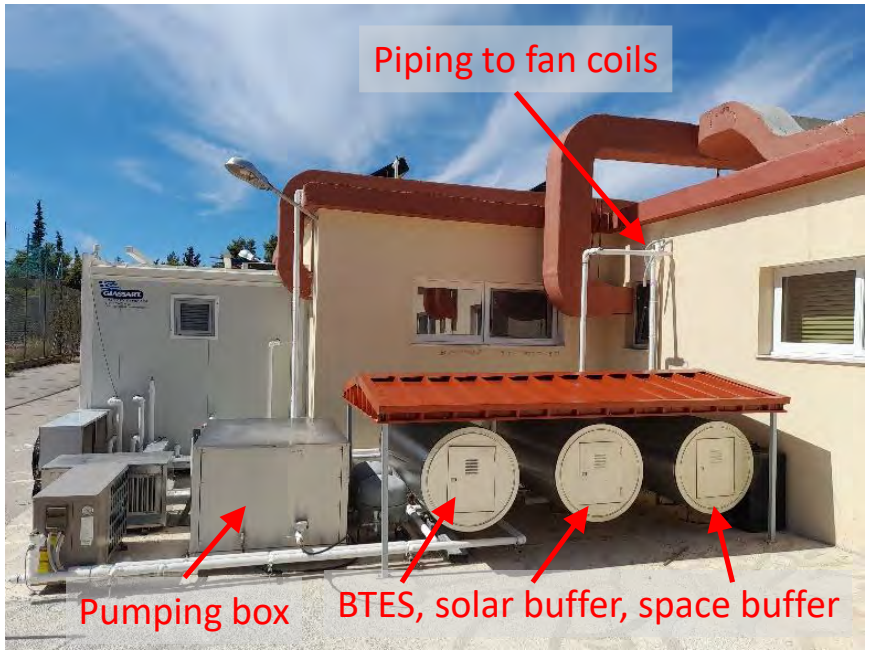
Renewables for clean energy in buildings in a future power system

- Innovative renewable energy-based technologies/solutions for **heating**, **cooling** & **energy storage**
  - Multi-source heat pump
  - Energy sources: PV/Thermal collectors
  - Energy storage: borehole thermal energy storage (BTES)
  - Control & flexibility: building energy management systems
- Test solutions in wide variety of buildings: new/renovated, different sizes, uses, locations

Flexibility from buildings - RES4BUILD

Pilot 1 - Office building in Athens





Indoor temp setpoints

- Day [08:00 to 21:00] : 20.5 - 21°C
- Night [21:00 to 08:00]: 18 - 19°C.

Component	Temperature boundaries
Solar buffer	Min: 0 °C, Max: 23 °C
DHW tank	Min: 10 °C, Max: 60 °C (bottom part of the tank)
	Min: 46 °C, Max: 51 °C (top part of the tank)
Space heating tank	Min: 40 °C, Max: 52 °C
BTES	Min: 14 °C, Max: 21 °C

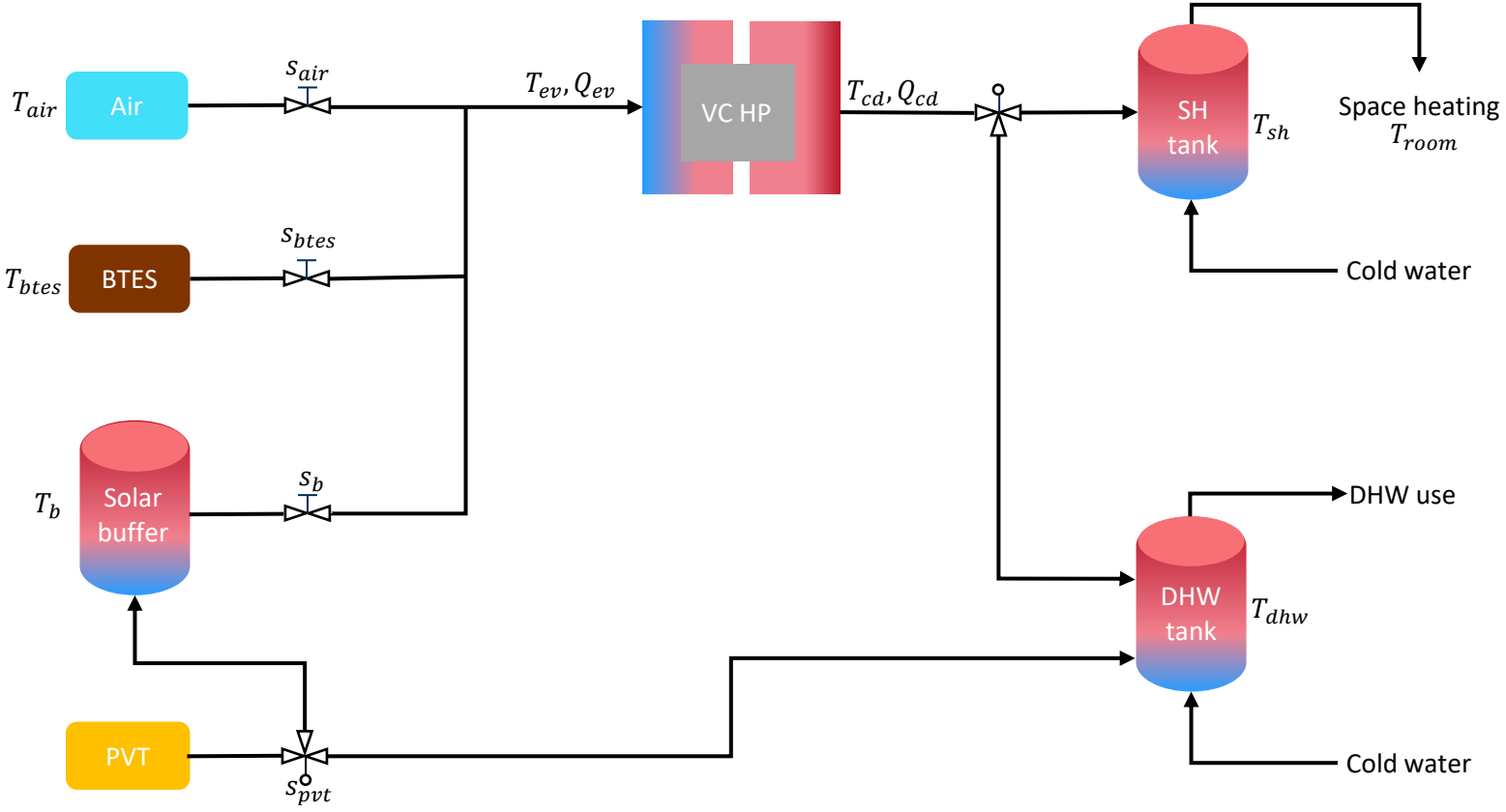
Flexibility from buildings - multi-source heat pumps

Pilot 1 - Office building in Athens

- Objective
  - Use system flexibility to operate the heat pump in an energy efficient manner
    - High coefficient of performance
- External inputs – forecasts
  - Domestic hot water & space heating demand
  - Outdoor temperature & solar irradiance

Flexibility from buildings - multi-source heat pumps

Pilot 1 - Office building in Athens



BTES: borehole thermal energy system

VC HP: Vapour compressor heat pump

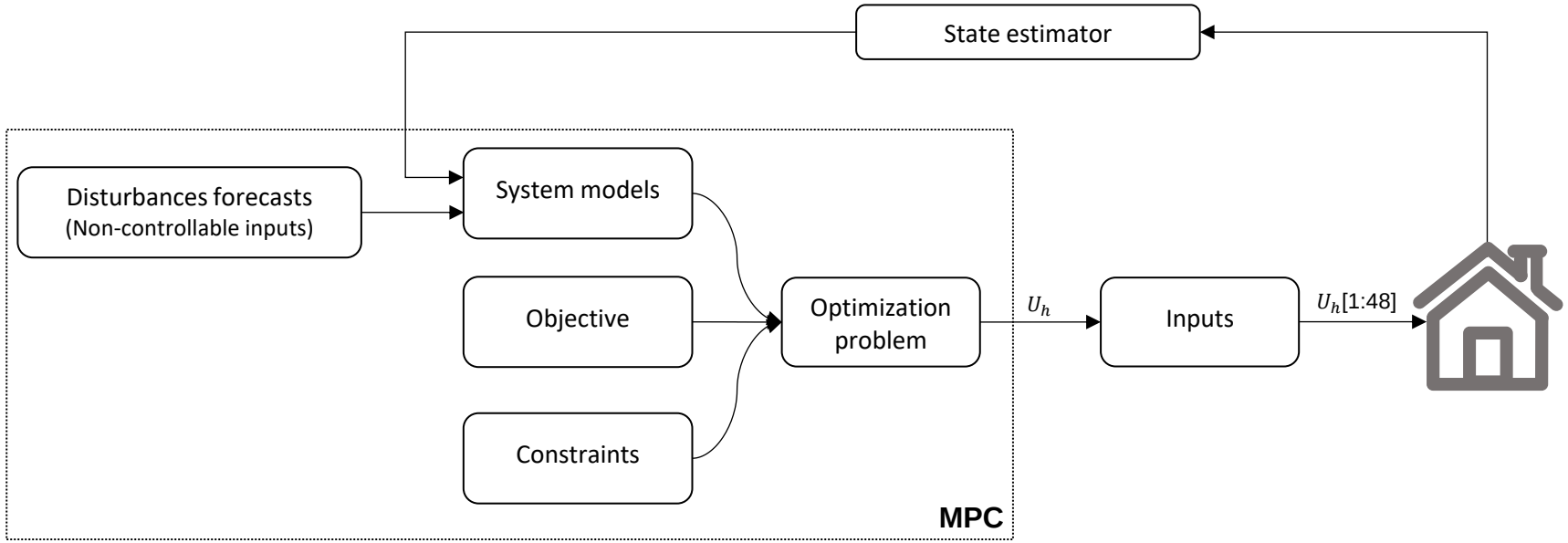
SH: space heating

DHW: domestic hot water

Flexibility from buildings - multi-source heat pumps

Pilot 1 - Office building in Athens

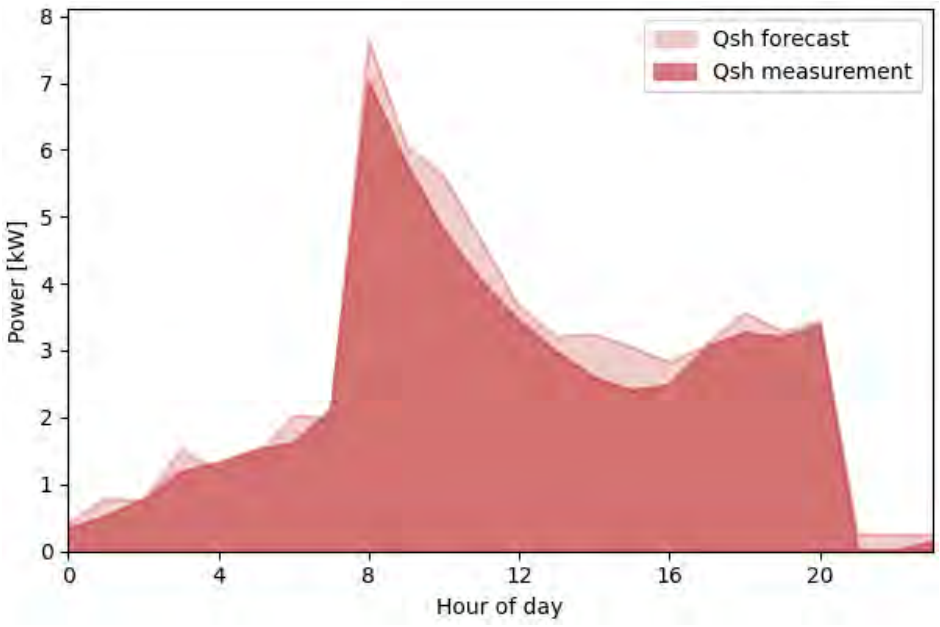
Optimisation problem solved using **model predictive control** (MPC)

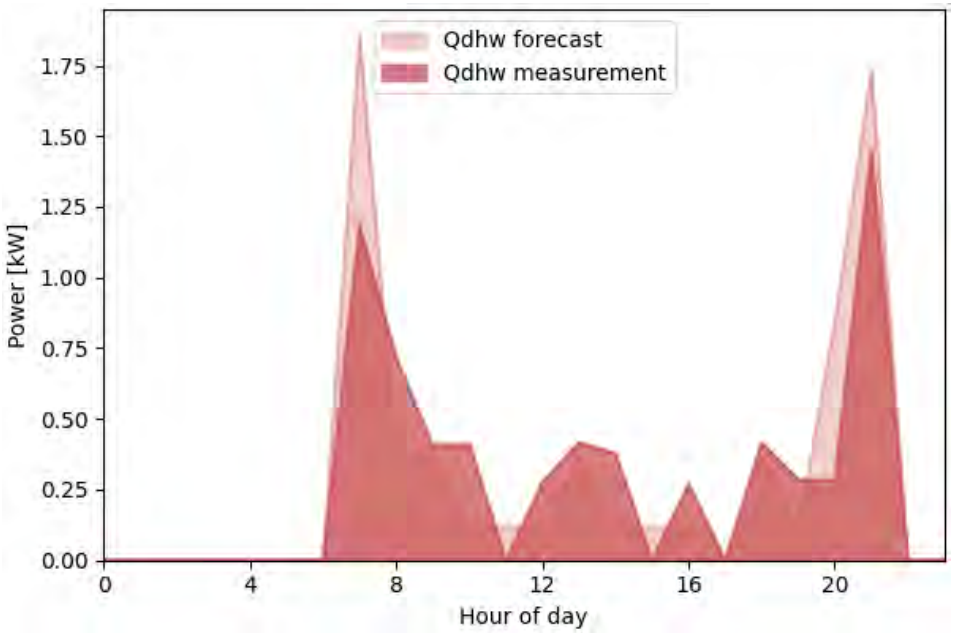


Flexibility from buildings - multi-source heat pumps

Pilot 1 - Office building in Athens

Real world implementation



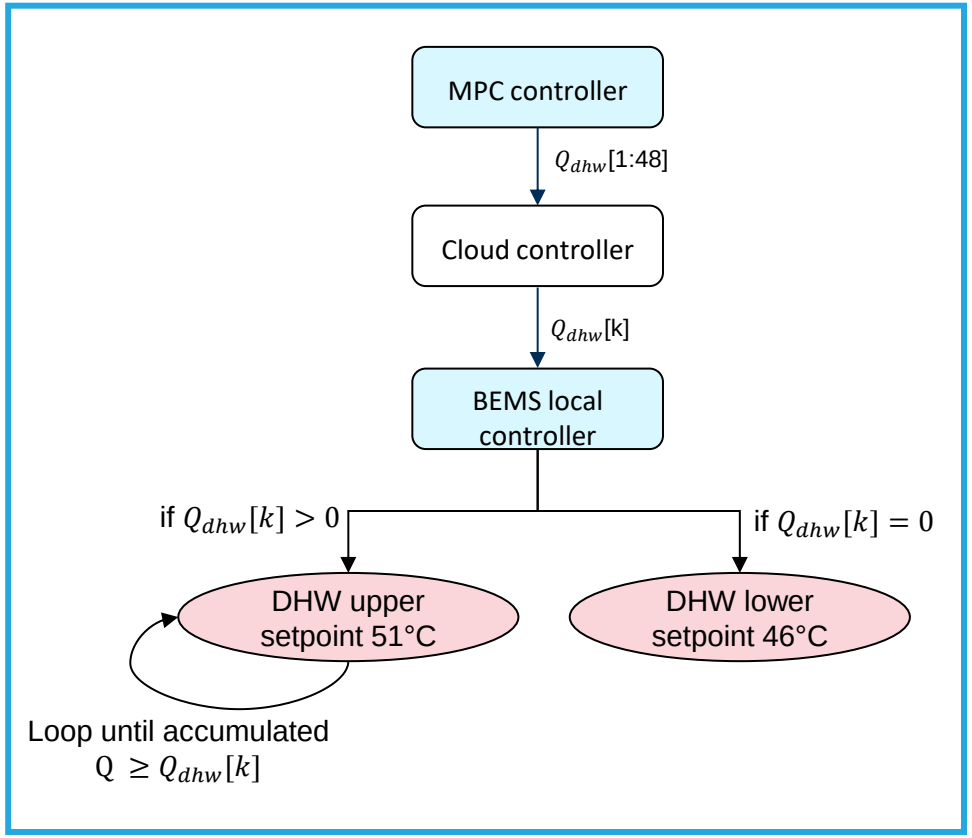


Flexibility from buildings - multi-source heat pumps

Pilot 1 - Office building in Athens

Real world implementation

- MPC output translated to temperature setpoints
  - Local BEMS – installed onsite



MPC: Model predictive control

BEMS: Building Energy management system

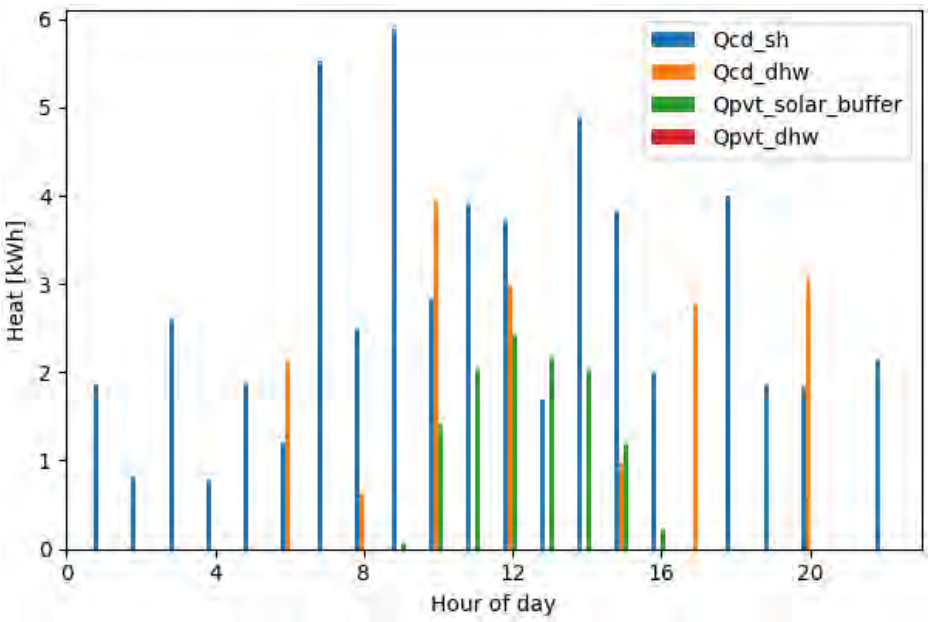
SH: space heating,

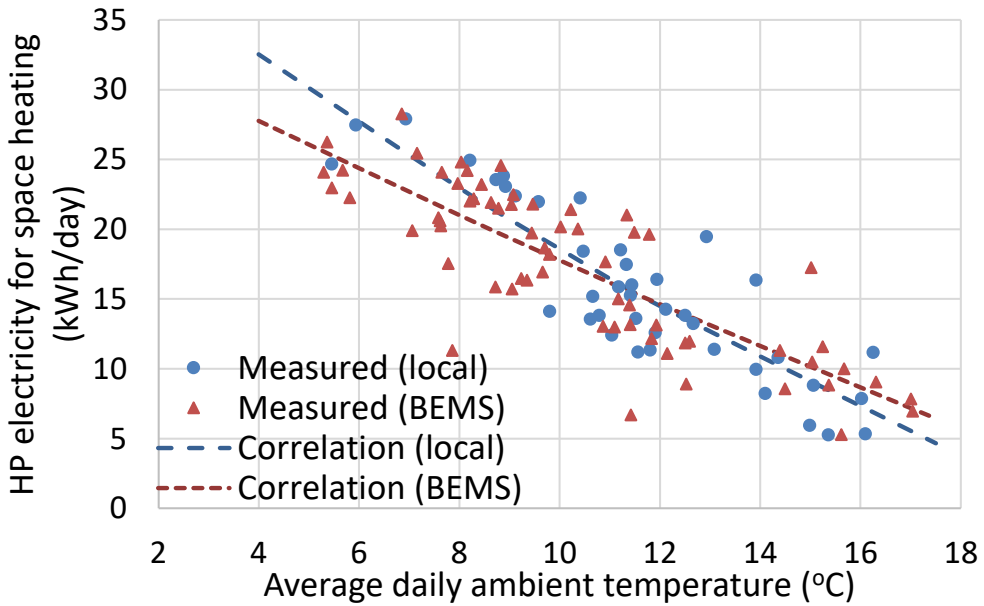
DHW: domestic hot water

Flexibility from buildings - multi-source heat pumps

Pilot 1 - Office building in Athens

Real world implementation





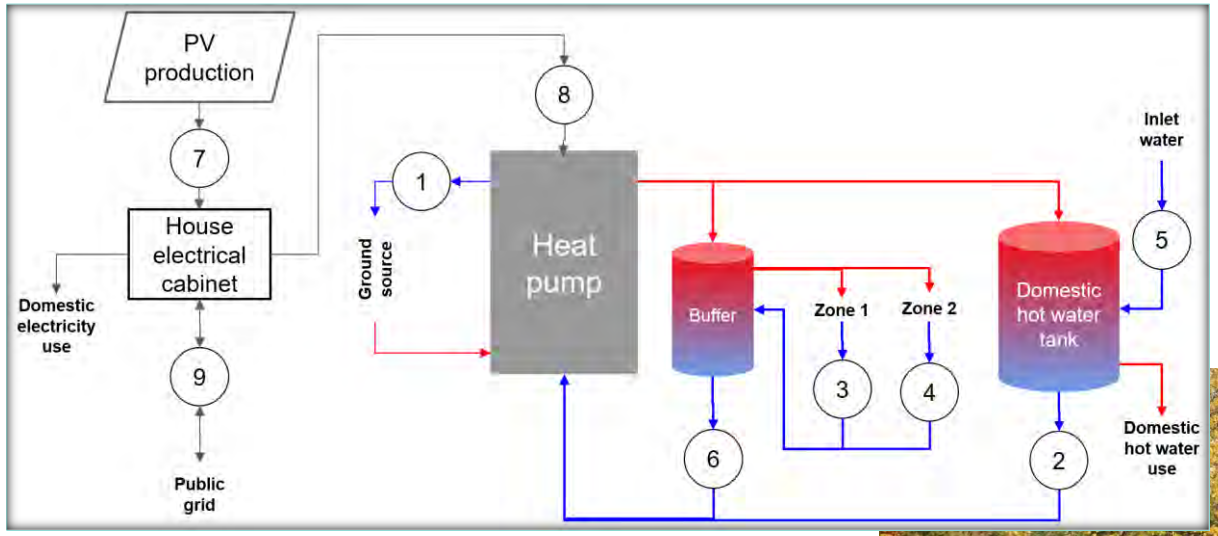
SH: space heating,
DHW: domestic hot water

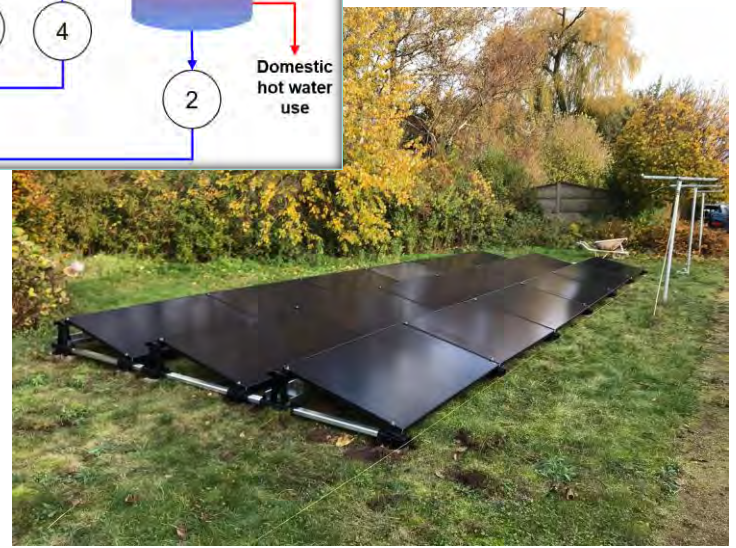
PVT: PV thermal collectors
BEMS: Building Energy management system

Qcd_sh: heat for space heating,
Qcd_dhw: heat for domestic hot water

Flexibility from buildings - hot water tank

Pilot 2 – Single family house Belgium



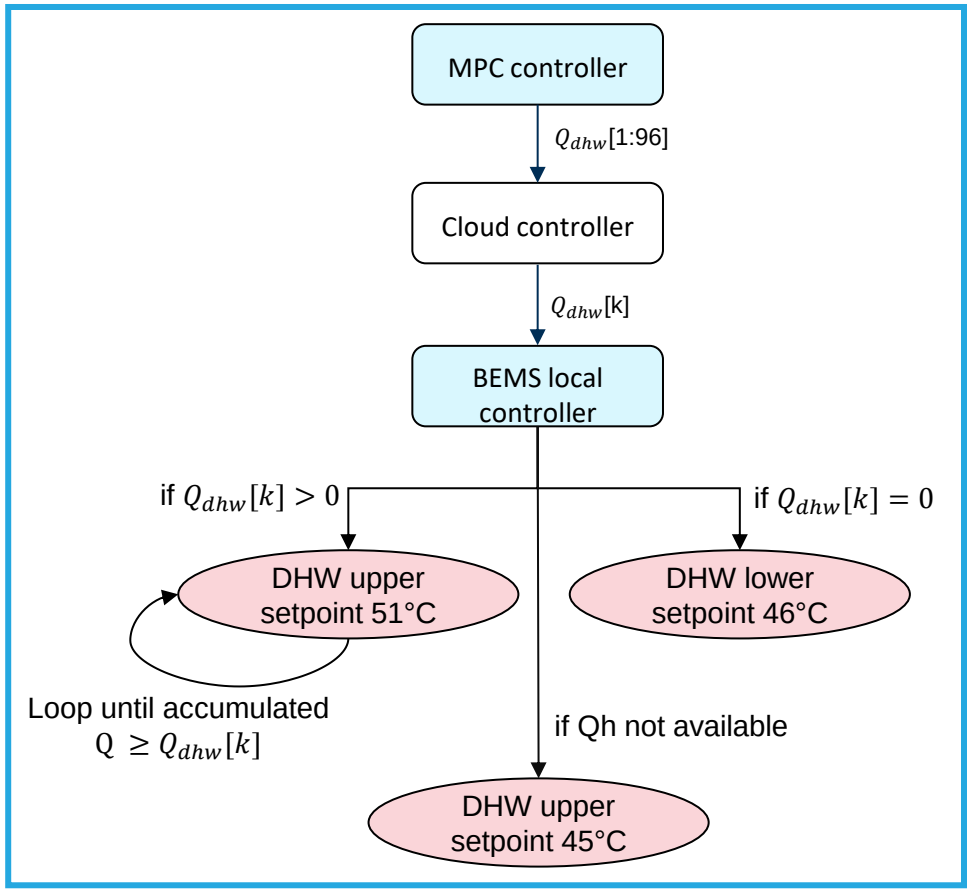


Flexibility from buildings - hot water tank

Pilot 2 - Single family house Belgium

Objective

- Maximisation of self-consumption of the PV production
- Commercial HP → only control the setpoint of DHW buffer
  - Thermal flexibility of house is not used
- DHW buffer setpoint
  - No heat production: 44 °C
  - Max heat production: 52 °C
  - No control setpoint: 45 °C



Flexibility from buildings - hot water tank

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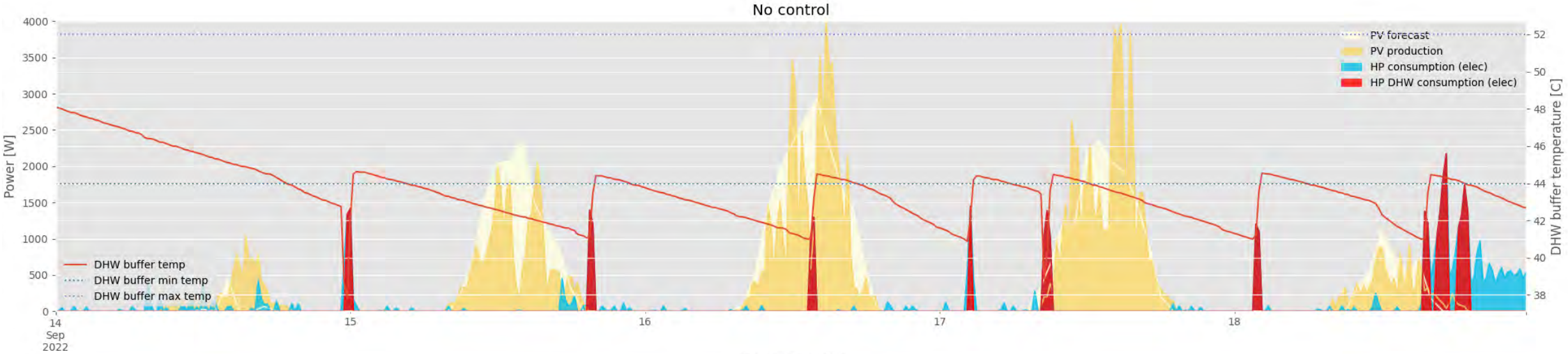

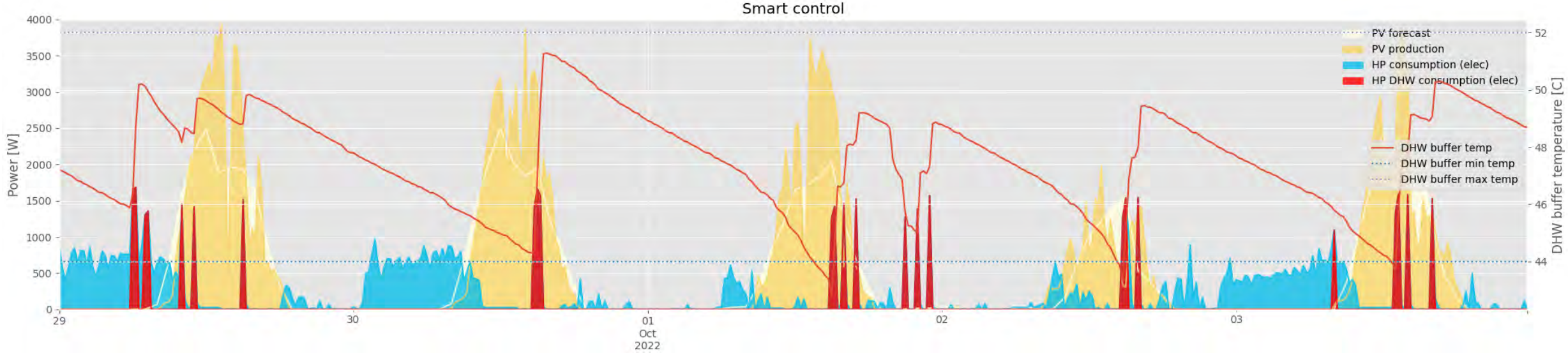

Flexibility from buildings

Conclusions & lessons learned

Improved self-consumption of local generation

Improved indoor comfort

Reduced electricity cost

Fall-back controller is essential during deployment & testing

Human angle is critical during deployment & testing

Interoperability is important

Thank you!

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